Lab 6: A/B Testing, Randomized Controlled Trials

Introduction

This A/B Testing report was performed on a clinical trial considering three individual treatments for pre-diabetics to reduce the progression to full diabetes. The three treatments recorded in our data - metformin (met), rosiglitazone (rosi), and lifestyle adjustments - and their corresponding outcomes were used in calculating cross tabulation, bootstrapping for each individual treatment's average and pairwise average treatment effects, and corresponding confidence intervals to determine whether each treatment should be adopted into practice.

Data Overview

Unnamed: 0	treatment	outcome
1	met	success
2	rosi	failure
3	rosi	success
4	lifestyle	success
5	met	success

Table 1. Head of data table used to perform calculations

Cross Tabulation and Average Outcomes

To begin testing, cross tabulation and the computation of the average outcomes for each treatment were computed.

Outcome Treatment	Failure	Success
lifestyle	109	125
met	120	112
rosi	90	143

Table 2. Cross tabulation of the three treatments and their outcomes

The average successful outcomes for **met, rosi,** and **lifestyle** are as follows: **48.28%, 61.38%,** and **53.42%**. From performing these calculations, there is evidence that rosi has the highest success rate, followed by lifestyle, and then met; however, this evidence alone is not strong enough to conclude that rosi, lifestyle changes, and met are effective treatments and should be adopted.

Bootstrapping: Treatment Outcomes and Pairwise Average Treatment Effects

To further test the individual treatments, bootstrapping was performed ten thousand times to estimate individual average outcomes for each treatment as well as pairwise average treatment effects. The pairings are as follows: met and rosi, met and lifestyle, rosi and lifestyle. These outcomes were then plotted as kernel density estimator plots (KDE) and empirical cumulative density function plots (ECDF). The following figures summarize the findings.

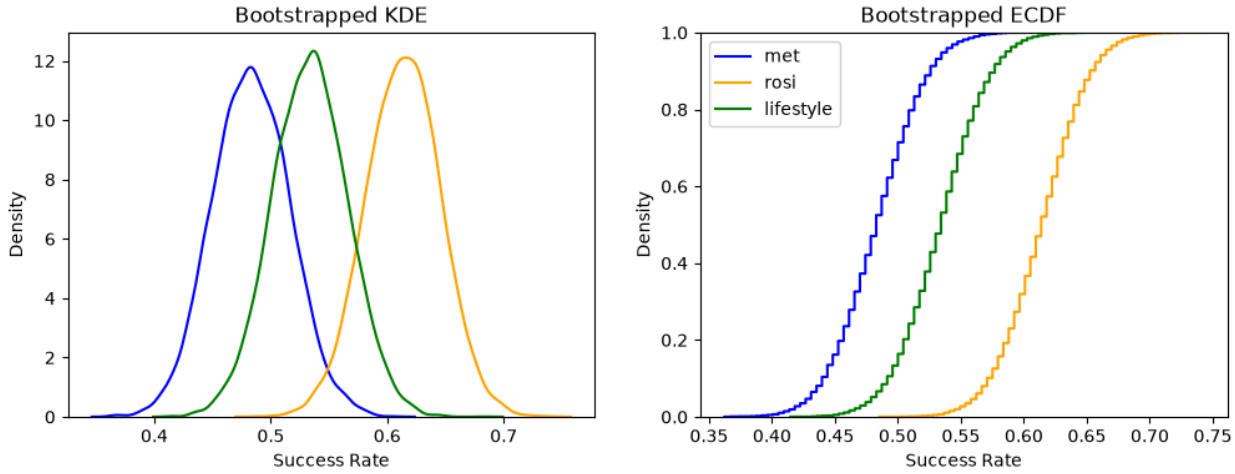


Fig 1. Bootstrapped KDE and ECDF plots for the average outcomes of met, rosi, and lifestyle treatments

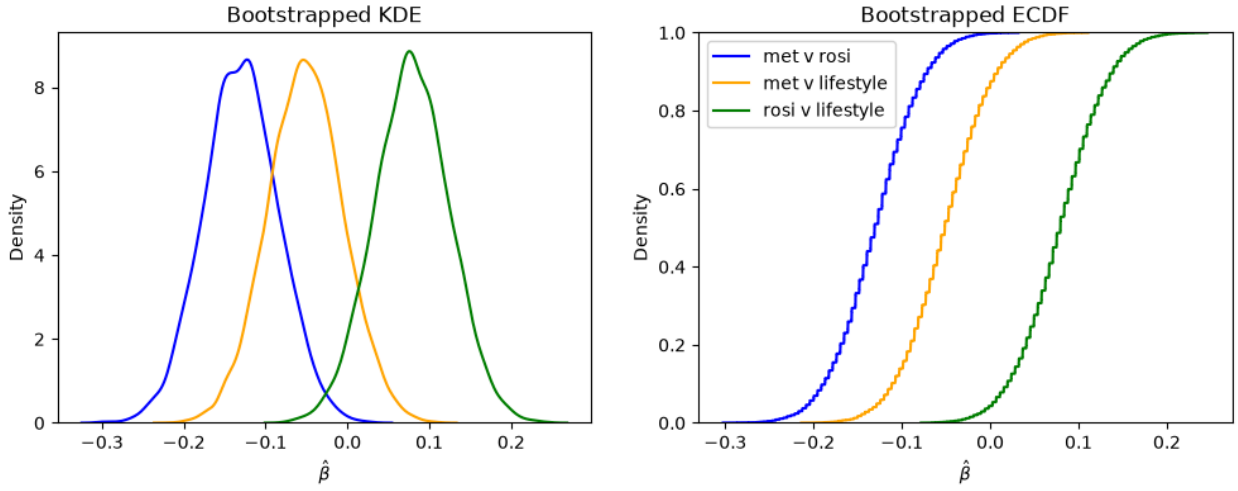


Fig 2. Bootstrapped KDE and ECDF plots for the pairwise average treatment effects.

As seen in figure 1, all three treatments follow similar normal bell curve shapes in their KDE and similarly shape steps for their ECDF plots. The rosi treatment has the highest success rate, followed by lifestyle, then met. These findings are also aligned with the findings in the cross tabulation and the overall average of outcomes.

Figure 2 plots the outcomes of the pairwise average treatment effects (ATE) calculated by bootstrapping. Using these plots, it is possible to determine evidence of differences between the success rates of treatments by evaluating the plotted curve's estimated range of betas. For the ATE of met and rosi, the beta outcomes consist primarily of negatives with few outcomes differing to zero. This suggests that there is evidence of met having a lower success rate than rosi with a few outcomes sharing no difference between the two treatments. The ATE of met and lifestyle consists of both negatives and positives as well as zero, so there is not consistent evidence that met outperforms lifestyle consistently and vice versa. Although, it is worth noting that most of the beta values are negative, indicating that met tends to have a lower success rate compared to lifestyle. Thirdly, the ATE of rosi and lifestyle consists of mostly positives and zero. While some outcomes show rosi and lifestyle having no differences in success rate, a majority of the estimates indicate rosi will have a higher success rate than a change of lifestyle. However, these plots do not provide evidence of the effectiveness of the treatments. Confidence Intervals must be calculated to determine statistically significant effects.

Confidence Intervals

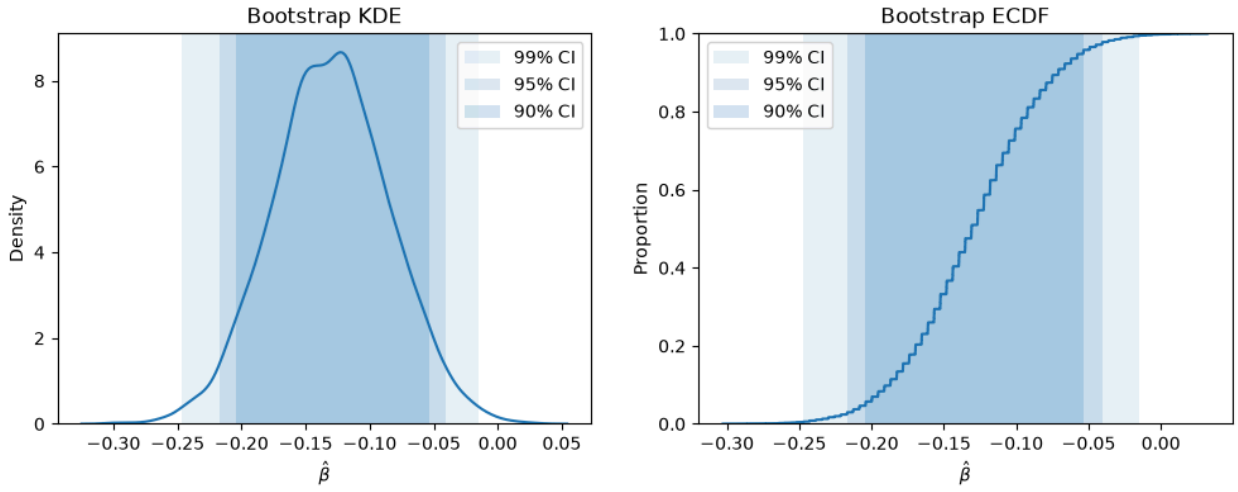
To test the effectiveness of each treatment, percentile bootstrapping was performed to calculate the confidence intervals for each pairwise ATE based on its ECDF. The results are as follows:

<u>ATE Met and Rosi</u>

CI90: [-0.20412165, -0.05362957]

CI95: [-0.21707165, -0.04064257]

CI99: [-0.2471142 , -0.01474397]



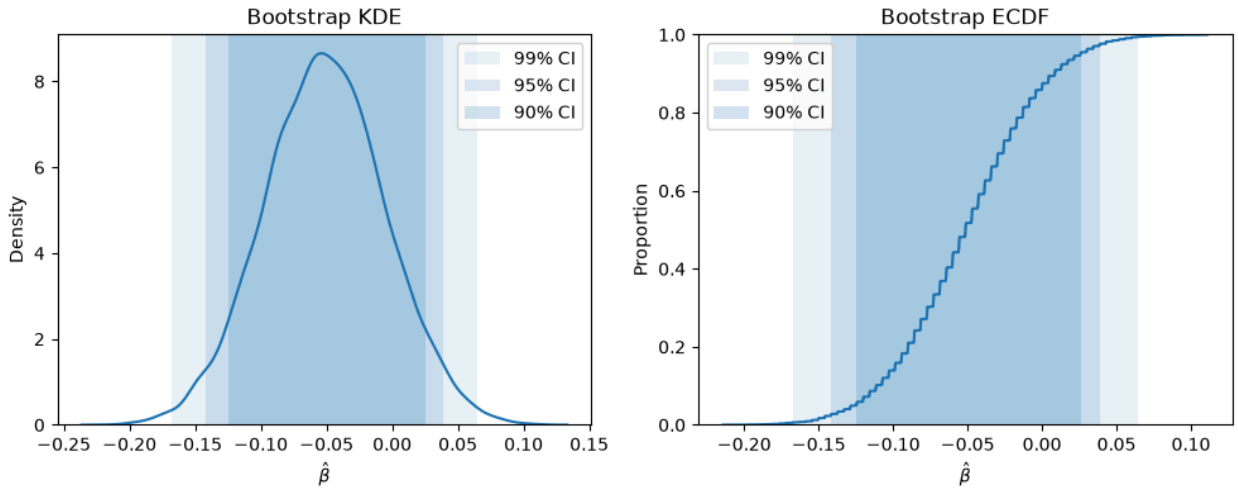
There is a statistically significant difference at all three confidence intervals. The confidence intervals at 90%, 95%, and 99% do not include zero, so we reject the null hypothesis that there are zero differences between the two treatments. There is evidence that rosi has a higher success rate than met.

ATE Met and Lifestyle

CI90: [-0.12466844, 0.02578839]

CI95: [-0.14157825 , 0.03879402]

CI99: [-0.16757018, 0.06474801]



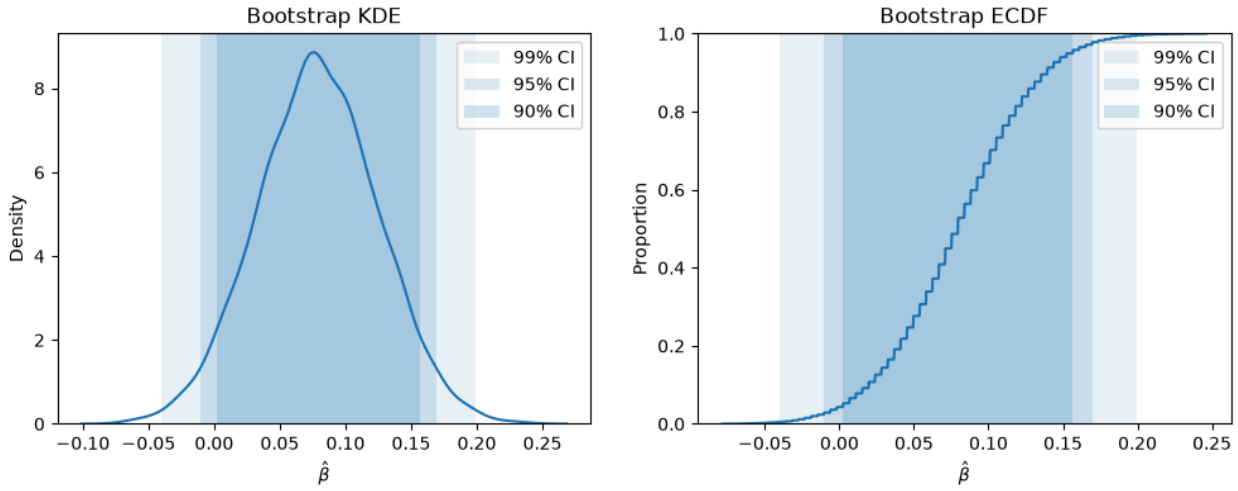
There is not a statistically significant difference between these two treatments. At the 90%, 95%, and 99% confidence intervals, zero is included. We fail to reject the null hypothesis that there are zero differences between the two treatments. There is not sufficient enough evidence to conclude that either treatment has a higher success rate than the other.

ATE Rosi and Lifestyle

CI90: [0.00251275, 0.15646895]

CI95: [-0.0104545, 0.16947287]

CI99: [-0.04035068, 0.19942409]



There is a statistically significant difference between the two treatments at the 90% confidence interval because the interval is all positive betas. We reject the null hypothesis that the difference between the two treatments is equal to zero. At the 95% and the 99% confidence intervals, there is not a statistically significant difference between the two treatments. Both intervals contain zero, so we fail to reject the null hypothesis.

Conclusion

After testing all three treatments based on their success rates, there is evidence that rosiglitazone has the highest success rate, and it would be the best treatment to adopt. When comparing rosiglitazone to metformin, there is a statistically significant difference between the two treatments indicating rosiglitazone to have a higher success rate than metformin at the 90%, 95%, and 99% confidence intervals. When comparing rosiglitazone to lifestyle changes, there is evidence that rosiglitazone has a higher success rate than lifestyle changes, but only at the 90% confidence interval, suggesting a weaker difference between the two treatments and statistical significance being dependent on the chosen confidence interval.